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Transfer-enhanced Federated Learning with Dynamic Clustering for Traffic Management in 5G Open RAN

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Abstract

User-centric traffic management in 5G Open RAN faces severe challenges due to multi-vendor heterogeneity, frequent handovers, and stringent QoS requirements. To address these issues, we propose TFL-CORAN, a transfer-enhanced federated reinforcement learning framework with dynamic soft clustering for scalable UE-level adaptation. In TFL-CORAN, UEs perform local DDQN-based learning and generate resource-control recommendations, gNBs enforce feasible slot-level scheduling and relay context/model updates, and the RIC coordinates clustering, aggregation, and personalized model redistribution. An offline-trained Variational Graph Auto-Encoder (VGAE) embeds heterogeneous UE contexts into a latent space, where Gaussian Mixture Model (GMM) memberships guide membership-aware federated aggregation. A transfer initialization mechanism further mitigates cold starts for new and handover UEs. Evaluations in realistic O-RAN scenarios show that TFL-CORAN improves QoS satisfaction, throughput, latency, reliability, and adaptation time over centralized, federated, and cluster-based baselines with manageable UE-side overhead.

Keywords: 5G Open RAN, Federated Learning, Transfer Learning, Dynamic Clustering, Resource Allocation

1. Introduction

The Open Radio Access Network (O-RAN) paradigm reshapes radio access control by disaggregating software and hardware functions and enabling multi-vendor interoperability [1, 2]. This openness is essential for flexible deployment, but it also makes traffic management more difficult because measurement, scheduling, and control functions are distributed across user equipments (UEs), next-generation NodeBs (gNBs), and RAN Intelligent Controller (RIC) functions. In practical O-RAN systems, a single controller rarely observes a synchronized global state across all UEs, cells, and vendors, while UE-level channel quality, interference, traffic demand, mobility, and service requirements change quickly due to handovers and mixed traffic classes such as eMBB, URLLC, and mMTC. As a result, policies optimized only at the cell aggregate can become misaligned with per-UE quality-of-service (QoS) objectives, especially for newly activated UEs and handover UEs that lack sufficient local experience. These characteristics motivate

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a UE-centric traffic-management framework that can learn from local feedback, coordinate across distributed O-RAN entities, and adapt under mobility-driven non-independent and identically distributed (non-IID) behavior.

Deep Reinforcement Learning (DRL) has been widely studied for radio resource allocation and traffic steering [3, 4, 5]. RL is suitable for O-RAN traffic management because resource control is inherently sequential: a scheduling-related decision in one slot affects later throughput, latency, reliability, queueing, and interference. Unlike supervised learning, optimal action labels are rarely available for every UE context. Moreover, deriving an accurate mathematical model for joint mobility, traffic demand, interference coupling, and scheduling outcomes is difficult in multi-cell O-RAN environments. Therefore, model-free RL is useful because it can improve resource-control behavior directly from interaction feedback without requiring a fully specified analytical model of the radio environment. However, centralized DRL is difficult to deploy directly in O-RAN because it requires timely global observations, repeated retraining under context drift, and additional mechanisms to capture intra-cell diversity among UEs with different service targets, mobility patterns, and interference regimes.

Federated Learning (FL) provides a natural way to coordinate distributed learning in such environments [6, 7]. Instead of collecting all UE trajectories at one controller, UEs can train local models and periodically upload model updates through their serving gNBs to a coordinator. Nevertheless, conventional FL is not sufficient by itself because many FL-based O-RAN approaches treat a gNB or a cell as one learning client, thereby overlooking UE-level heterogeneity inside the cell. Naive averaging across heterogeneous UEs can also mix incompatible model updates when users experience different mobility, interference, and traffic regimes. Cluster-based FL mitigates this issue by grouping similar clients [8, 9], but fixed or hard assignments are fragile under continuous mobility and handovers. Graph-based grouping methods can capture relational similarity among users or cells [10, 11], yet fully online graph reconstruction and graph-model training can be too expensive for frequent O-RAN control updates. Transfer learning has also been explored for knowledge reuse across cells or tasks [12, 10], but existing approaches often operate at coarse cell-level granularity and do not tightly couple transfer with clustering-aware federated aggregation for newly activated and handover UEs.

To address these limitations, we propose TFL-CORAN, a transfer-enhanced federated reinforcement learning framework with dynamic clustering for traffic management in 5G O-RAN. The central idea is to align the learning architecture with O-RAN operation: UEs learn from local QoS feedback and generate resource-control recommendations, gNBs enforce feasible scheduling decisions, and the RIC coordinates model aggregation and context-aware grouping. At a slower structural time scale, the RIC refreshes UE groupings so that aggregation reflects mobility and traffic drift without requiring online graph retraining at every slot. This time-scale separation is used as an operational design matching computation and communication requirements, rather than as a formal convergence guarantee.

TFL-CORAN realizes this design through three connected mechanisms. First, an offline-trained Variational Graph Auto-Encoder (VGAE) [13] maps UE context reports into latent embeddings that capture both UE-level features and similarity structure. Second, a Gaussian Mixture Model (GMM) [14] infers soft memberships in this latent space; these memberships directly determine how local model updates are weighted and how cluster models are mixed into UE-personalized models. Third, a transfer initialization mechanism uses the same latent-context similarity to warm-start newly activated and handover UEs from relevant neighboring or cluster models. Thus, representation learning, soft clustering, federated aggregation, and transfer initialization are coupled around the same UE-context structure.

Our contributions are summarized as follows:

- We formulate UE-centric O-RAN traffic management as a federated reinforcement learning problem with explicit decision boundaries among UEs, gNBs, and the RIC. The formulation distinguishes UE-generated recommendations from gNB-enforced scheduling decisions and links the resulting QoS outcomes to the RL reward.

- We propose TFL-CORAN, which combines offline VGAE representation learning [13], online GMM soft clustering [14], and membership-aware federated aggregation to provide UE-level personalization under mobility-driven non-IID conditions.

- We introduce a transfer initialization mechanism for newly activated and handover UEs, using latent-context similarity to reuse parameters from relevant neighboring or cluster models and reduce post-activation and post-handover adaptation delay.

- We evaluate TFL-CORAN against centralized, federated, and cluster-based baselines using realistic O-RAN simulation settings, reporting QoS satisfaction, throughput, latency, reliability, adaptation time, learning behavior, and computational/communication complexity.

2. Related Works

O-RAN resource allocation and traffic steering must operate under fragmented observability, tight control-loop budgets, fast UE mobility, and heterogeneous service requirements across eMBB, URLLC, and mMTC slices. Accordingly, prior work can be organized by who learns and acts (central controller, gNB, RIC, or UE), who enforces the resulting decision (gNB scheduler or RIC-side control function), and how heterogeneity is handled (federation, clustering, transfer, or representation learning). Graph-based representation learning and embedding methods [15, 11] suggest that relational structure can encode UE similarity beyond simple metric distances, including dynamic settings such as temporal graphs [16]. However, in O-RAN traffic management, such representations must be coupled to practical control loops and aggregation rules rather than treated as standalone clustering tools.

2.1. Centralized DRL for O-RAN Control

Centralized DRL typically places the learning agent at a cell controller, gNB-side scheduler, or RIC function, assuming sufficiently rich observations to optimize slice-level objectives such as throughput–latency trade-offs in network slicing [17]. Choi et al. [3] studied dynamic bandwidth allocation under high mobility using a DRL policy enhanced with temporal prediction, showing that forecasting can improve decision-making when channel and load fluctuate. Kavehmadavani et al. [18, 4] combined traffic prediction with DRL-based traffic steering, highlighting the benefit of anticipating near-future demand in beyond-5G/6G O-RAN settings. Filali et al. [5] adopted a hierarchical O-RAN view with near-RT and non-RT control loops, demonstrating that multi-level DRL can improve URLLC-oriented latency and slice isolation. Recent O-RAN slicing studies also emphasize that latency-aware control must be mapped carefully to O-RAN components and control-loop timing [19]. These works motivate RL because radio control is sequential and current scheduling decisions affect later delay, reliability, queueing, and interference. Nevertheless, centralized DRL often requires consistent global observations and centralized retraining, and cell-level policies may require additional personalization mechanisms when UE contexts inside a cell are highly diverse.

2.2. Federated DRL for UE-Centric Learning

Federated learning [6, 7] shifts training to distributed participants by letting local models be updated independently and coordinated through periodic aggregation. In federated DRL for O-RAN, the learning agent can be instantiated at UEs or edge-side entities, while a coordinator aggregates model updates to improve network-wide behavior. Abouaomar et al. [12] applied federated DRL to O-RAN slicing, showing that decentralized policy learning can improve eMBB throughput while maintaining URLLC latency constraints. Ndikumana et al. [20] combined FL with deep Q-learning for joint task offloading and routing decisions in O-RAN, reporting reduced end-to-end delay under distributed optimization. Singh and Nguyen [21] focused on communication-efficient FL in O-RAN using compression and momentum-style updates. Policy distillation has also emerged as an alternative way to transfer knowledge among heterogeneous policies in network slicing [22]. However, distillation requires exchanging teacher outputs or action distributions rather than parameter updates, and most existing FDRL formulations still have limited mechanisms to track mobility-driven non-independent and identically distributed (non-IID) drift or to prevent negative transfer when UE contexts diverge rapidly.

2.3. Clustering-Enhanced Federated Learning under Heterogeneity

To mitigate non-IID effects and reduce aggregation variance, cluster-based federated schemes group clients and perform within-cluster training or aggregation. Mughal et al. [9] proposed intra-cluster FL for heterogeneous networks, showing that grouping similar participants can improve convergence behavior relative to a single global aggregate. Zhang et al. [10] developed a UE-centric traffic steering framework that incorporates knowledge reuse and selective aggregation, illustrating that update weighting or pruning can improve scalability when client relevance varies. Yet, many cluster-based approaches rely on fixed or hard assignments, which are difficult to maintain under continuous UE movement and handover events. Moreover, clustering is often treated as an auxiliary preprocessing step, rather than being directly connected to the aggregation weight and the personalized model delivered to each UE. This motivates soft-membership aggregation, where cluster information continuously affects both who contributes to a model and which model is redistributed.

2.4. Dynamic and Graph-Aware Client Grouping

Dynamic clustering attempts to update group structure as client distributions drift. FedVar [23] adjusts client influence in aggregation and triggers clustering in heterogeneous regimes when client updates become inconsistent. FedGCD [8] leverages graph-based community detection, using learned representations and modularity-driven partitioning to identify groups with similar data characteristics. Graph embedding methods [15, 11] further support the idea that relational information can capture similarity patterns that are not visible from isolated UE features. Nevertheless, fully online graph reconstruction and graph-model training can be computationally heavy for frequent O-RAN control updates. For this reason, a practical UE-centric framework should separate offline representation learning from lightweight online inference and explicitly define how the resulting embeddings affect clustering, aggregation, and personalization.

2.5. Transfer Learning, Reward Design, and Cold Starts

Transfer learning has been explored to reuse knowledge across cells or tasks to improve adaptation speed. Zhang et al. [10] demonstrated that UE-centric steering can benefit from knowledge transfer when a UE enters a new context, while Tang and Jiang [24] discussed transfer-assisted FL as a general approach to accelerate convergence. Adaptive reward designs have also been studied for related cell activation and network-control problems [25], suggesting that reward shaping can help when dominant service violations change over time. In this work, we use fixed service-normalized reward terms for controlled comparison across baselines and evaluate hard QoS satisfaction separately so that latency or reliability violations are not hidden by other reward components. Existing O-RAN-oriented transfer designs often operate at coarse cell-level granularity and do not directly distinguish newly activated UEs from handover UEs. They also rarely couple transfer initialization with the same latent representation and soft memberships used for federated aggregation.

2.6. Summary of Gaps and Motivation

In summary, centralized DRL provides adaptive sequential decision-making but depends on rich global observations and often lacks explicit UE-level personalization. Federated DRL improves distributed learning, but naive aggregation can be fragile under mobility-driven non-IID drift. Clustering and graph-aware grouping mitigate heterogeneity, but hard assignments and fully online graph learning are difficult to align with practical O-RAN control-loop constraints. Transfer learning can reduce cold-start delay, but prior work typically treats it separately from clustering-aware aggregation and does not clearly specify the UE/gNB/RIC decision boundaries. These limitations motivate TFL-CORAN, which combines offline graph-based representation learning, online GMM soft memberships, membership-aware federated aggregation, and transfer initialization in a single UE-centric O-RAN traffic-management framework.

3. Proposed Method

3.1. O-RAN-Aligned Framework Overview

Figures 1 and 2 summarize the proposed TFL-CORAN framework and its operating time scales. Figure 1 shows how the learning functions are mapped onto O-RAN entities, while Figure 2 shows when each function is executed. The main purpose of these figures is to clarify that TFL-CORAN is not a simple concatenation of DDQN, VGAE, GMM, FL, and transfer learning. Instead, these components form a single closed-loop traffic-management system in which each output is used by a subsequent control or learning operation.

At the slot level, each UE observes its local radio and service state and generates a resource-control recommendation. The recommendation is not directly enforced by the UE. The serving gNB collects recommendations from its associated UEs, combines them with channel state, service priority, and resource constraints, and then applies the feasible scheduling decision. The resulting throughput, latency, and reliability are measured at the UE/gNB side and returned as QoS feedback for local DDQN training. Thus, the UE learns from the QoS outcome induced by the actual gNB-enforced scheduling decision.

At the federated-learning level, UEs do not send replay samples or raw trajectories to the RIC. After several local DDQN episodes, each UE uploads a model update through its serving gNB. The gNB only relays model updates and context summaries; it does not perform FL aggregation. The RIC-side coordinator aggregates UE updates and

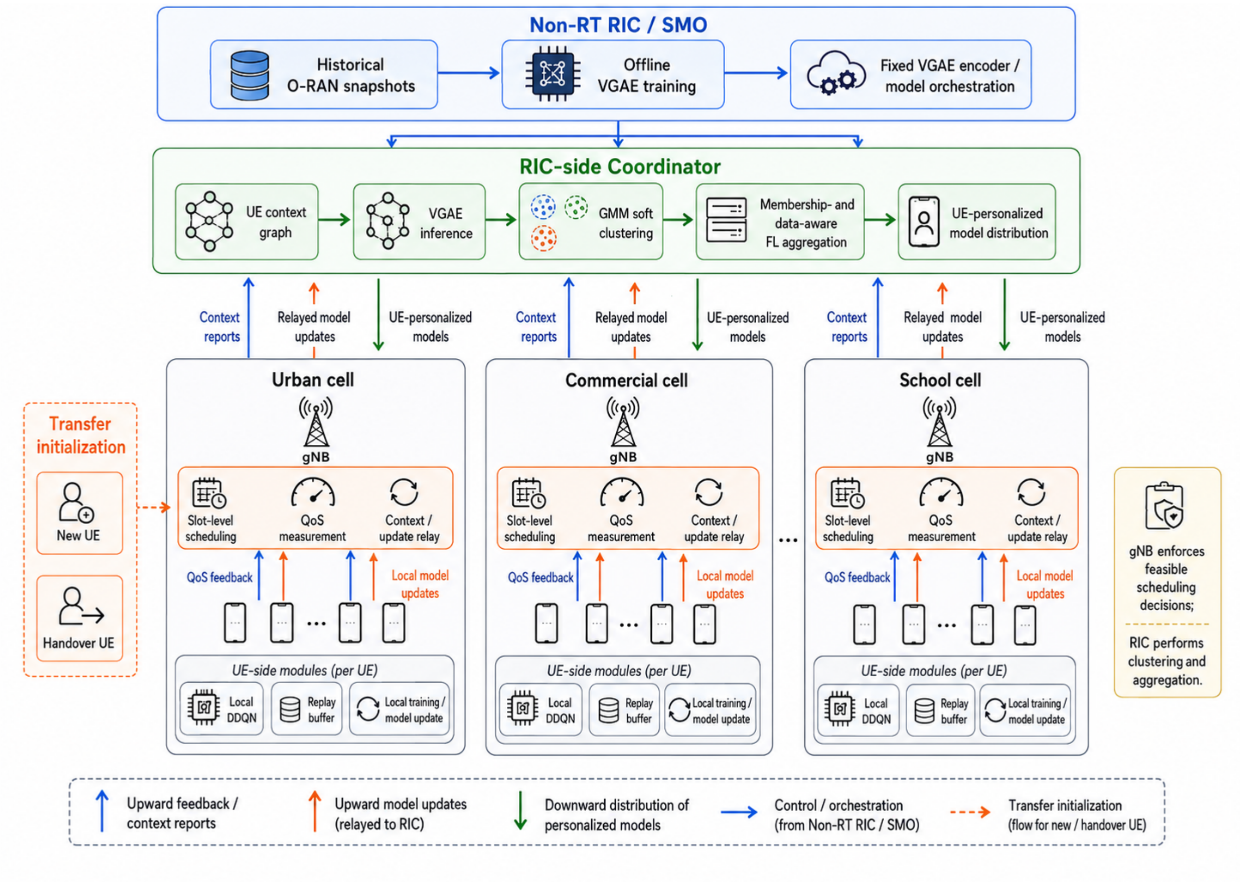

Figure 1: Proposed framework of TFL-CORAN. UEs perform local DDQN inference/training and generate resource-control recommendations; gNBs enforce feasible slot-level scheduling decisions, measure QoS outcomes, and relay context/model updates; the RIC-side coordinator performs VGAE inference, GMM soft clustering, membership- and data-aware FL aggregation, and UE-personalized model distribution. Offline VGAE training and model orchestration are handled at the Non-RT RIC/SMO side.

redistributes personalized models. To prevent updates from dissimilar UEs from being averaged uniformly, the RIC first organizes UEs into soft context groups using VGAE-based embeddings and GMM memberships.

At the clustering-refresh level, the RIC updates the UE similarity structure less frequently than the slot-level DDQN interaction. This slower refresh is used because graph construction, VGAE inference, and GMM fitting describe structural context drift rather than instantaneous channel fluctuation. The time-scale separation is therefore an operational design that matches each task to its computational and communication cost. We do not use it as a formal convergence guarantee; empirical convergence behavior is evaluated in Section 5.

The full loop proceeds as follows. First, the Non-RT RIC/SMO trains a VGAE encoder offline using historical O-RAN snapshots. Second, at a clustering refresh, the RIC-side coordinator constructs a UE similarity graph from context reports and obtains latent UE embeddings using the fixed VGAE encoder. Third, the RIC fits a GMM in the latent space and obtains soft memberships. Fourth, during each FL round, these memberships determine how UE model updates are weighted and how cluster models are mixed into UE-personalized models. Finally, for new and handover UEs, the same latent-context structure is used to initialize the UE from a contextually relevant model.

3.2. System Model, Observability, and Time Scales

We consider a multi-cell O-RAN deployment with multiple gNBs and a RIC-side coordinator. Each gNB consists of radio units (RUs) and distributed units (DUs), serves a set of UEs, and performs slot-level scheduling. Let $\mathcal{G}_{\mathrm{NB}}$ denote the set of gNBs, and let $\mathcal{U}^{(r)} = \{u_1, \ldots, u_{U_r}\}$ denote the active UE set at FL round r. The active set may evolve

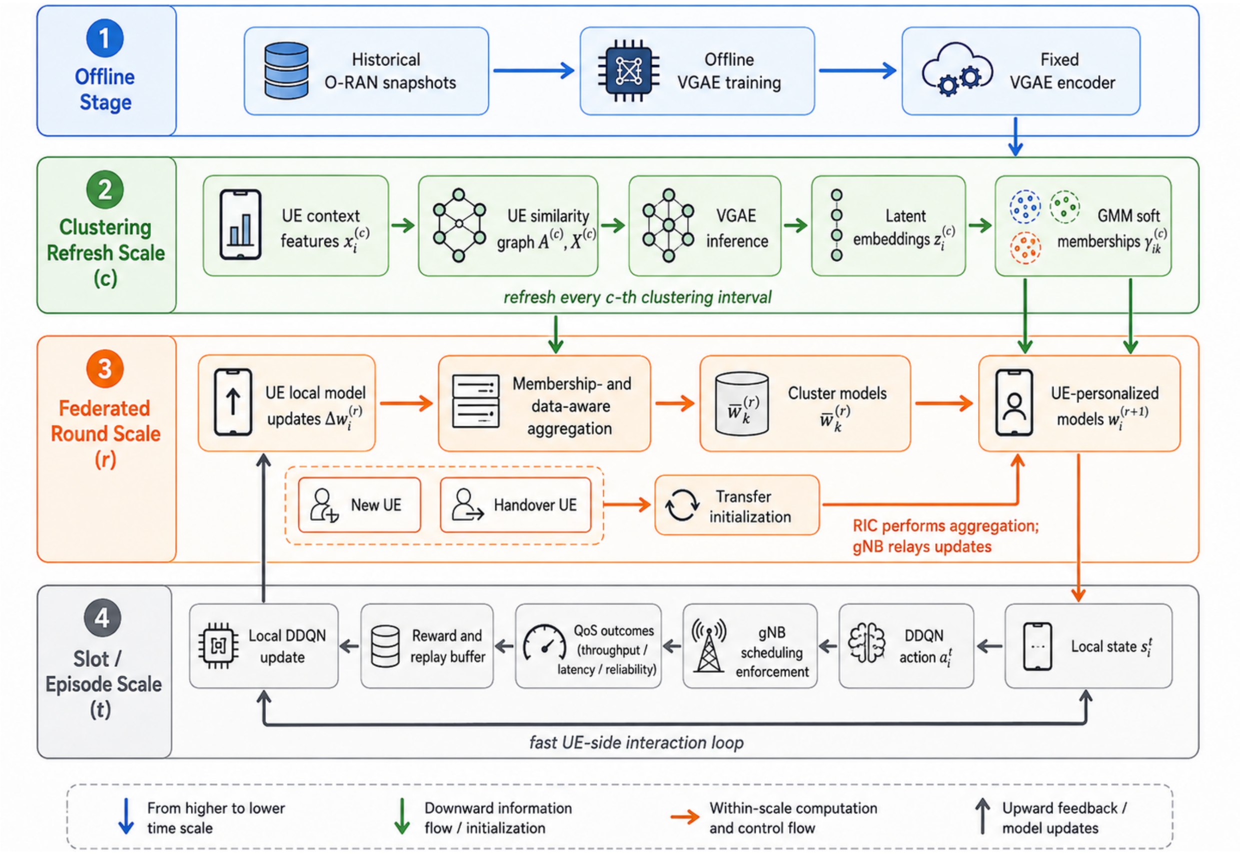

Figure 2: Operational pipeline and time-scale hierarchy of TFL-CORAN. Offline VGAE training produces a fixed encoder. At each clustering refresh, the RIC constructs a UE similarity graph, infers latent embeddings, and updates GMM memberships. During each FL round, UE updates are aggregated using membership and local-experience weights to produce cluster models and UE-personalized models. At the slot/episode scale, UEs perform local DDQN learning while gNBs enforce feasible scheduling decisions.

because newly activated UEs join the system and existing UEs hand over across cells. The serving gNB of UE u_i at slot t is denoted by $g_i^t \in \mathcal{G}_{\mathrm{NB}}$.

Each UE requests one service class

$$\kappa_i \in \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}, \tag{1}$$

with a service-specific QoS target tuple

$$\mathbf{q}_i = (\Theta_i^{\mathrm{tar}}, L_i^{\mathrm{tar}}, \eta_i^{\mathrm{tar}}), \tag{2}$$

where Θ_i^{tar}, L_i^{tar}, and η_i^{tar} are the target throughput, latency, and reliability, respectively.

The observability of each entity is different. A UE can observe its local service state, previous QoS feedback, and limited radio indicators received from its serving gNB. A gNB can observe scheduling-related information for its associated UEs, such as channel quality, contention on subbands, queue/backlog state, and QoS outcomes after scheduling. The RIC does not execute slot-level scheduling. Instead, it receives two types of summarized information through gNBs: model updates for FL and context reports for clustering.

A context report is defined as a slowly refreshed summary of UE operating conditions. At clustering refresh index c, UE u_i is represented by

$$x_i^{(c)} = [S_i^{(c)}, I_i^{(c)}, T_i^{(c)}, M_i^{(c)}], \tag{3}$$

where $S_i^{(c)}$, $I_i^{(c)}$, $T_i^{(c)}$, and $M_i^{(c)}$ denote signal strength, interference, traffic load, and mobility intensity, respectively. These quantities are obtained by averaging or summarizing slot-level measurements over the most recent reporting

Table 1: O-RAN Roles, Observations, Outputs, and Time Scales in TFL-CORAN

Entity	Observable information	Main operation and output	Time scale
UE	Local state, previous QoS feedback, service class	DDQN inference/training; resource-control recommendation; local model update	Slot / episode / FL round
gNB	UE recommendations, channel quality, contention, QoS outcomes	Feasible scheduling decision; QoS measurement; relay of context reports and model updates	Slot / FL round
RIC-side coordinator	Context reports and relayed model updates	VGAE inference, GMM clustering, membership-aware aggregation, UE-personalized model distribution	Clustering refresh / FL round
Non-RT RIC/SMO	Historical O-RAN snapshots	Offline VGAE training and fixed encoder orchestration	Offline

window before refresh c. Thus, $x_i^{(c)}$ is not a raw trajectory but a compact context summary used for representation learning and clustering.

The time indices used in TFL-CORAN are separated as follows. The slot index t denotes the fastest scheduling and DDQN interaction time. A local episode e consists of L slots. An FL round r occurs every τ_E local episodes, after which UEs upload local model updates. A clustering refresh index c is updated every τ_C FL rounds. For a given FL round r, the latest available clustering index is denoted by

$$c(r) = \left\lfloor \frac{r}{\tau_C} \right\rfloor. \tag{4}$$

The GMM memberships computed at refresh $c(r)$ are reused for all FL rounds until the next clustering refresh.

Table 1 summarizes the role, observation, output, and time scale of each O-RAN entity. Table 2 summarizes the main notation used in the following subsections.

3.3. UE-Level DDQN Formulation and gNB Scheduling Feedback

RL is used because O-RAN traffic management is a sequential decision problem. A resource recommendation made at one slot affects not only the current resource allocation but also subsequent throughput, latency, reliability, queueing, and interference conditions. Supervised learning would require optimal labels for each UE context, which are generally unavailable. Static optimization would require an accurate and constantly updated mathematical model of mobility, traffic demand, interference coupling, and scheduler response, which is difficult to obtain in heterogeneous O-RAN deployments. Therefore, each UE learns a local policy from its own interaction feedback.

At slot t, UE u_i observes the local state

$$s_i^t = [P_i^t, d_i^t, \ell_i^{t-1}, \eta_i^{t-1}, \theta_i^{t-1}, \kappa_i], \tag{5}$$

where P_i^t is mobility context, d_i^t is a data-rate or buffer-related indicator, and $(\ell_i^{t-1}, \eta_i^{t-1}, \theta_i^{t-1})$ are the most recently observed latency, reliability, and throughput. The UE then outputs a recommendation

$$a_i^t = [a_i^{(f)}, a_i^{(\rho)}, a_i^{(p)}], \tag{6}$$

where $a_i^{(f)} \in \mathcal{F}$ is the preferred subband, $a_i^{(\rho)} \in \mathcal{R}$ is the preferred rate level, and $a_i^{(p)} \in \mathcal{P}$ is the priority parameter.

The actual resource decision is made by the gNB. For the UE set $\mathcal{U}_g^t$ associated with gNB g at slot t, the gNB applies a scheduling function

$$b_i^t = \Phi_g\left(\{a_j^t : j \in \mathcal{U}_g^t\}, \chi_g^t\right), \qquad u_i \in \mathcal{U}_g^t, \tag{7}$$

In our implementation, $\Phi_g(\cdot)$ is realized by the slot-level scheduler in the simulation. If a UE recommendation is feasible and no conflicting request has higher scheduling utility, the gNB applies the recommended subband/rate/priority. If multiple UEs compete for the same subband or if the recommendation violates resource constraints, the gNB resolves the contention using service priority, predicted MCS/rate, and current channel conditions. Therefore, the applied decision b_i^t may differ from the original UE recommendation a_i^t under congestion.

Table 2: Main Notation

Symbol	**Description**	**Symbol**	**Description**
t, e, r, c	Slot, episode, FL round, and clustering refresh indices	L, τ_E, τ_C	Slots per episode, episodes per FL round, and FL rounds per clustering refresh
$\mathcal{U}^{(r)}$	Active UE set at FL round r	g_i^t	Serving gNB of UE u_i at slot t
κ_i	Service class of UE u_i	$\Theta_i^{\text{tar}}, L_i^{\text{tar}}, \eta_i^{\text{tar}}$	Target throughput, latency, and reliability
s_i^t	Local DDQN state of UE u_i	a_i^t	UE-generated resource-control recommendation
b_i^t	gNB-enforced scheduling decision for UE u_i	$\Phi_g(\cdot)$	gNB scheduling function
$\theta_i^t, \ell_i^t, \eta_i^t$	Throughput, latency, and reliability after scheduling	R_i^t, H_i^t	Reward and hard QoS satisfaction indicator
$w_i^{(r)}, w_{i,\text{loc}}^{(r)}$	Received UE model and locally trained model	$\Delta w_i^{(r)}, n_i^{(r)}$	Uploaded model update and local transition count
$x_i^{(c)}, \tilde{x}_i^{(c)}$	Context report and normalized context vector	$A_{ij}^{(c)}$	Weighted graph similarity between UEs u_i and u_j
$h_i^{(\ell)}, z_i^{(c)}$	GCN hidden representation and VGAE latent embedding	$\gamma_{ik}^{(c)}$	GMM membership probability of UE u_i in cluster k
$\bar{w}_k^{(r)}$	Cluster-k model at FL round r	δ	Transfer fusion coefficient for handover UEs

where b_i^t is the feasible scheduling decision applied to UE u_i, and χ_g^t denotes the gNB-side scheduling context, including channel quality, available subbands, predicted MCS, service priorities, and contention state. This formulation resolves the distinction between the UE recommendation and the final scheduling action. The UE controls a_i^t, but the environment response is generated through the gNB-enforced decision b_i^t.

After applying b_i^t, the UE/gNB observes the resulting QoS metrics

$$\theta_i^t = \theta_i^t(b_i^t), \qquad \ell_i^t = \ell_i^t(b_i^t), \qquad \eta_i^t = \eta_i^t(b_i^t). \tag{8}$$

The immediate reward assigned to the UE recommendation a_i^t is computed from the QoS outcome induced by b_i^t:

$$R_i^t(s_i^t, a_i^t, b_i^t) = \frac{\eta_i^t(b_i^t)}{\eta_i^{\text{tar}}} + \frac{\theta_i^t(b_i^t)}{\Theta_i^{\text{tar}}} - \frac{\ell_i^t(b_i^t)}{L_i^{\text{tar}}}. \tag{9}$$

Thus, the reward is action-dependent through the mapping $a_i^t \to b_i^t \to (\theta_i^t, \ell_i^t, \eta_i^t)$. Because each QoS term is normalized by its service-specific target, throughput or reliability is less likely to dominate the reward solely due to its numerical scale. The DDQN therefore learns which recommendations tend to produce favorable feasible scheduling outcomes after the gNB applies its constraints.

For evaluation, QoS satisfaction is not computed by the additive reward. Instead, it is treated as a hard service condition:

$$H_i^t = \mathbf{1}\left[\theta_i^t \geq \Theta_i^{\text{tar}}, \ell_i^t \leq L_i^{\text{tar}}, \eta_i^t \geq \eta_i^{\text{tar}}\right]. \tag{10}$$

Although we use equal coefficients for the normalized reward terms to ensure a controlled comparison across baselines, a weighted or adaptive reward design could further emphasize the currently dominant SLA violation.

Therefore, high throughput cannot hide a latency or reliability violation in the reported QoS satisfaction metric.

The long-term policy objective is

$$\max_{\pi} \sum_{u_i \in \mathcal{U}^{(r)}} \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma_{\text{RL}}^t R_i^t(s_i^t, a_i^t, b_i^t)\right], \tag{11}$$

where $\gamma_{\mathrm{RL}} \in (0,1)$ is the RL discount factor. This objective defines what the policy aims to optimize, while the DDQN loss below defines how each UE updates its local model.

UE u_i maintains a DDQN action-value function $Q_i(s,a;w_i^{(r)})$, where $w_i^{(r)}$ is the online-network parameter at FL round r. The target-network parameter is denoted by $(w_i^-)^{(r)}$. Given a transition $(s_i^t, a_i^t, R_i^t, s_i^{t+1})$, DDQN selects the next action using the online network:

$$a_i^{t+1,*} = \arg\max_{a'} Q_i(s_i^{t+1}, a'; w_i^{(r)}), \tag{12}$$

and evaluates that action using the target network:

$$y_i^t = R_i^t + \gamma_{\mathrm{RL}} Q_i(s_i^{t+1}, a_i^{t+1,*}; (w_i^-)^{(r)}). \tag{13}$$

The online parameter is updated by minimizing the temporal-difference loss

$$\mathcal{L}_i(w_i^{(r)}) = \left(y_i^t - Q_i(s_i^t, a_i^t; w_i^{(r)})\right)^2, \tag{14}$$

with learning rate α_{lr}:

$$w_i^{(r)} \leftarrow w_i^{(r)} - \alpha_{\mathrm{lr}} \nabla_{w_i^{(r)}} \mathcal{L}_i(w_i^{(r)}). \tag{15}$$

The target network is synchronized periodically.

3.4. Federated Reinforcement Learning and Model Updates

Local DDQN allows each UE to react to fast radio and traffic changes, but local learning alone is inefficient when UEs have limited experiences or when new and handover UEs appear. Federated reinforcement learning is used to coordinate these local policies without requiring a centralized learner to observe every slot-level transition. The RIC performs coordination at the FL-round scale, while UE/gNB interaction continues at the slot scale.

At the beginning of FL round r, UE u_i receives a model $w_i^{(r)}$ from the RIC. This model may be a personalized mixture of cluster models or a transfer-initialized model for a new/handover UE, as described later. The UE then performs local DDQN updates for τ_E episodes. Let $w_{i,\mathrm{loc}}^{(r)}$ denote the resulting locally trained parameter after these updates. The uploaded model update is

$$\Delta w_i^{(r)} = w_{i,\mathrm{loc}}^{(r)} - w_i^{(r)}. \tag{16}$$

Let $n_i^{(r)}$ denote the number of local transitions used by UE u_i during round r. The pair $(\Delta w_i^{(r)}, n_i^{(r)})$ is relayed by the serving gNB to the RIC. The gNB does not aggregate model parameters.

Uniform FedAvg is sensitive to O-RAN heterogeneity because UEs with very different service classes, interference regimes, or mobility patterns may produce conflicting updates. For this reason, TFL-CORAN uses context-aware clustering before aggregation. The next two subsections define how the RIC constructs context embeddings and soft memberships, and Section 3.7 explains how these memberships are used in FL aggregation.

3.5. Context-Aware Representation Learning via VGAE

The RIC needs a compact representation that tells which UEs are operating under similar conditions. Directly clustering raw measurements is fragile because signal strength, interference, traffic load, and mobility have different scales and may be correlated through cell topology. TFL-CORAN therefore uses a VGAE to embed UE context reports into a latent space that captures both node features and pairwise similarity structure. The VGAE is trained offline at the Non-RT RIC/SMO using historical O-RAN snapshots, and the fixed encoder is used online by the RIC-side coordinator. Here, offline training means that the VGAE parameters are learned before online FL operation from previously collected or simulated O-RAN context-graph snapshots. During online control, these VGAE parameters are frozen; the RIC only performs forward inference to obtain UE embeddings at clustering refresh intervals. Thus, online operation does not include gradient-based VGAE retraining.

Let

$$X^{(c)} = [x_1^{(c)}, \ldots, x_{U_r}^{(c)}]^\top \tag{17}$$

be the matrix of context reports at clustering refresh c. Each feature dimension is normalized using mean and standard deviation computed from the offline training snapshots:

$$\tilde{x}_i^{(c)} = \frac{x_i^{(c)} - \mu_x}{\sigma_x + \epsilon_x}, \tag{18}$$

where μ_x and σ_x are feature-wise statistics, and ϵ_x is a small constant.

The RIC constructs a continuous weighted similarity graph

$$\mathcal{G}^{(c)} = (\mathcal{V}^{(c)}, \mathcal{E}^{(c)}, A^{(c)}), \tag{19}$$

where each node corresponds to a UE. The edge weight between UEs u_i and u_j is

$$A_{ij}^{(c)} = \frac{1}{1 + \|\tilde{x}_i^{(c)} - \tilde{x}_j^{(c)}\|_2^2}. \tag{20}$$

Here, $A_{ij}^{(c)} \in (0, 1]$ is a soft similarity value, not a binary link indicator. A large value means that two UEs have similar signal, interference, traffic, and mobility contexts during refresh window c.

The VGAE encoder is implemented with graph convolutional layers. At layer ℓ, the hidden representation of UE u_i is

$$h_i^{(\ell)} = \rho\left(W^{(\ell)} \sum_{j \in \mathcal{N}(i)} \frac{A_{ij}^{(c)}}{\sqrt{d_i d_j}} h_j^{(\ell-1)}\right), \qquad h_i^{(0)} = \tilde{x}_i^{(c)}. \tag{21}$$

Here, $W^{(\ell)}$ is the trainable weight matrix of layer ℓ, $\rho(\cdot)$ is a nonlinear activation function, $\mathcal{N}(i)$ is the neighbor set of UE u_i, and $d_i = \sum_j A_{ij}^{(c)}$ is the weighted degree. This weighted aggregation allows the embedding of a UE to reflect both its own context and the contexts of nearby or similar UEs.

The VGAE parameterizes a Gaussian posterior using a log-variance convention:

$$\mu_i = W_\mu h_i^{(L)}, \qquad \log \sigma_i^2 = W_\sigma h_i^{(L)}. \tag{22}$$

The latent embedding is sampled by the reparameterization trick:

$$z_i^{(c)} = \mu_i + \exp\left(\frac{1}{2} \log \sigma_i^2\right) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{23}$$

The vector $z_i^{(c)}$ is the latent context representation used by GMM clustering and transfer initialization.

Because the graph is weighted, the decoder reconstructs the continuous similarity score:

$$\hat{A}_{ij}^{(c)} = \sigma_{\text{sig}}\left((z_i^{(c)})^\top z_j^{(c)}\right), \tag{24}$$

where $\sigma_{\text{sig}}(x) = 1/(1 + \exp(-x))$ is the logistic sigmoid. The reconstruction loss uses the continuous adjacency values as soft labels:

$$\mathcal{L}_{\text{VGAE}} = -\sum_{i,j} \left[A_{ij}^{(c)} \log \hat{A}_{ij}^{(c)} + (1 - A_{ij}^{(c)}) \log(1 - \hat{A}_{ij}^{(c)})\right] + \text{KL}\left(q(z|x) \| \mathcal{N}(0, I)\right). \tag{25}$$

After offline training, only the encoder is used online. Thus, the online RIC-side cost is a forward pass through the fixed encoder rather than repeated graph-model training.

In the implementation, the latent dimension is set to $d_z = 32$ to balance representation capacity and geometric robustness. A larger latent space may increase representation capacity but can make similarity measurement less reliable due to high-dimensional distance concentration, whereas a smaller latent space may underrepresent heterogeneous UE contexts.

3.6. Dynamic Soft Clustering via GMM

The VGAE provides latent context embeddings, but the RIC still needs a way to decide which UE updates should influence one another during aggregation. Hard clustering is problematic under mobility because a UE near a cluster boundary may abruptly switch clusters after a small context change. TFL-CORAN therefore uses GMM-based soft clustering, so each UE can partially belong to multiple latent context regimes.

At clustering refresh c, the RIC fits a K-component GMM to the embeddings $\{z_i^{(c)}\}$:

$$p(z_i^{(c)}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(z_i^{(c)} | m_k, \Sigma_k), \tag{26}$$

where π_k is the mixture weight, m_k is the mean vector, and Σ_k is the covariance matrix of component k. Each Gaussian component represents a latent context regime, such as UEs with similar mobility, traffic load, and interference conditions. The components are not assumed to correspond one-to-one to service classes.

The GMM is estimated with EM. In the E-step, the RIC computes the posterior membership probability

$$\gamma_{ik}^{(c)} = \frac{\pi_k \mathcal{N}(z_i^{(c)} | m_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(z_i^{(c)} | m_j, \Sigma_j)}. \tag{27}$$

The value $\gamma_{ik}^{(c)} \in [0, 1]$ is the degree to which UE u_i belongs to component k, and $\sum_k \gamma_{ik}^{(c)} = 1$.

In the M-step, the effective component size, mixture weight, mean, and covariance are updated as

$$N_k = \sum_i \gamma_{ik}^{(c)}, \qquad \pi_k = \frac{N_k}{|\mathcal{U}^{(r)}|}, \tag{28}$$

$$m_k = \frac{1}{N_k} \sum_i \gamma_{ik}^{(c)} z_i^{(c)}, \tag{29}$$

$$\Sigma_k = \frac{1}{N_k} \sum_i \gamma_{ik}^{(c)} (z_i^{(c)} - m_k)(z_i^{(c)} - m_k)^\top + \epsilon_\Sigma I, \tag{30}$$

where ϵ_Σ is a small diagonal regularization parameter. The first refresh uses k-means initialization in the latent space, while later refreshes warm-start EM from the previous GMM parameters. EM terminates when the log-likelihood improvement is below a threshold or when the maximum number of iterations is reached.

The membership vector

$$\boldsymbol{\gamma}_i^{(c)} = [\gamma_{i1}^{(c)}, \ldots, \gamma_{iK}^{(c)}] \tag{31}$$

is held fixed between clustering refreshes. For FL round r, the RIC uses the latest membership vector from $c(r)$. These memberships directly determine aggregation weights and UE-personalized model synthesis in the next subsection.

3.7. Membership- and Experience-Aware Federated Aggregation

The purpose of aggregation is not to produce a single model that ignores UE heterogeneity. Instead, the RIC uses GMM memberships to construct cluster models and then mixes them into UE-personalized models. This design allows UEs in similar latent contexts to reinforce one another while reducing negative transfer from dissimilar UEs.

A remaining issue is that UEs may collect different amounts of local experience within the same FL round. If aggregation uses only $\gamma_{ik}^{(c)}$, a UE with very few replay samples may receive excessive weight. Therefore, TFL-CORAN combines soft membership and local experience count. For FL round r and latest clustering index $c(r)$, the effective contribution weight of UE u_i to cluster k is

$$\beta_{ik}^{(r)} = \gamma_{ik}^{(c(r))} n_i^{(r)}. \tag{32}$$

The cluster-level update is

$$\Delta \bar{w}_k^{(r)} = \frac{\sum_{u_i \in \mathcal{U}^{(r)}} \beta_{ik}^{(r)} \Delta w_i^{(r)}}{\sum_{u_i \in \mathcal{U}^{(r)}} \beta_{ik}^{(r)}}. \tag{33}$$

The cluster model is updated as

$$\bar{w}_k^{(r)} = \bar{w}_k^{(r-1)} + \Delta\bar{w}_k^{(r)}. \tag{34}$$

Here, $\bar{w}_k^{(r)}$ is the model associated with latent context component k after FL round r.

Personalization occurs when the RIC synthesizes a UE-specific model from the cluster models:

$$w_i^{(r+1)} = \sum_{k=1}^{K} \gamma_{ik}^{(c(r))} \bar{w}_k^{(r)}. \tag{35}$$

Thus, GMM memberships are used twice. First, they determine how each UE update contributes to each cluster model. Second, they determine how cluster models are mixed for each UE. If one membership dominates, the method behaves similarly to cluster-specific FL. If memberships are spread across clusters, the model becomes a smooth mixture, which is useful for boundary UEs and handover UEs. If memberships are uniform, the update approaches FedAvg-style aggregation.

3.8. Transfer-Based Initialization for New and Handover UEs

New and handover UEs are particularly vulnerable because their local DDQN models may be poorly matched to the current cell. A newly activated UE has no previous local model. A handover UE has a previous-cell model, but that model may not match the target cell's interference, traffic load, and scheduling conditions. TFL-CORAN uses the same latent context space learned by the VGAE to select a suitable initialization.

For UE u_i, the RIC first obtains $z_i^{(c(r))}$ using the fixed VGAE encoder. Let $\mathcal{U}_{g(i)}^{(r)}$ denote the active UE set served by the target gNB of u_i at FL round r. The most similar active UE is selected by cosine similarity in the latent space:

$$j^* = \arg \max_{j \in \mathcal{U}_{g(i)}^{(r)} \setminus \{i\}} \frac{(z_i^{(c(r))})^\top z_j^{(c(r))}}{\|z_i^{(c(r))}\|_2 \|z_j^{(c(r))}\|_2}. \tag{36}$$

Self-matching is excluded. Latent-space similarity is used because the VGAE embedding already incorporates normalized context features and graph-structural similarity. Cosine similarity is adopted because UEs with similar structural neighborhood patterns tend to have aligned embedding directions, even when their absolute embedding magnitudes differ due to asymmetric connectedness or local density variations. This makes angular similarity more robust than raw Euclidean distance for nearest-neighbor transfer in the learned latent space.

Latent-space similarity is used because the VGAE embedding already incorporates normalized context features and graph-structural similarity.

For a newly activated UE, there is no previous model. The UE is initialized from the most similar active UE's personalized model:

$$w_i^{(r+1)} = w_{j^*}^{(r+1)}. \tag{37}$$

If no suitable neighbor exists, the RIC falls back to the membership-weighted model in (35).

For a handover UE, the RIC fuses the previous-cell model and the target-context model:

$$w_i^{(r+1)} = \delta w_{i,\text{prev}} + (1-\delta) w_{j^*}^{(r+1)}, \qquad \delta \in [0, 1]. \tag{38}$$

The coefficient δ controls how much previous-cell knowledge is preserved after handover. A small δ gives more weight to the target-cell context, while a large δ preserves more of the UE's previous model. The value of δ is selected through the sensitivity analysis in Section 5. After initialization, both new and handover UEs enter the same local DDQN and FL aggregation loop as other active UEs.

4. Simulation Settings

4.1. Simulation Environment and Traffic Modeling

We evaluate TFL-CORAN in a realistically modeled O–RAN environment by integrating UERANSIM for RAN protocol emulation [26], Open5GS for core network functionality [27], and QuaDRiGa for geometry-based stochastic

Table 3: DDQN Parameter Settings

Parameter	Value
Learning rate α_{lr}	10^{-3}
Discount factor γ_{RL}	0.99
Batch size	64
Target network update	every 10 episodes
Replay buffer size	100,000
Exploration strategy	ϵ-greedy (1.0 $\rightarrow$ 0.01)
Optimizer	Adam

channel modeling [28, 29]. QuaDRiGa generates time-varying channel and SINR traces under UE mobility, which are then used by the slot-level scheduler to determine modulation and coding scheme (MCS), achievable transport block size, block error rate (BLER), and HARQ-like retransmission outcomes. UERANSIM and Open5GS emulate the RAN/core protocol stack, while the proposed learning modules operate on the resulting UE context reports, QoS measurements, and model updates.

The simulation starts with an initial active population of 375 UEs: 150 in the urban cell, 120 in the commercial cell, and 105 in the school cell. At each FL round, new UEs corresponding to 3% of the initial population are activated and added to the active UE set. In parallel, 10% of the currently active UEs are reassigned across cells to emulate handovers. No deactivation is applied in the main scenario; therefore, the active UE population gradually increases over rounds. All reported performance metrics are first computed over the active UE set at each round and then averaged across rounds and random seeds.

Each gNB operates at 3.5 GHz with 100 MHz bandwidth and manages $N_f = 6$ subbands. Multiple UEs may request the same subband in a slot, and the gNB scheduler resolves contention using the UE resource-control recommendations, service priorities, predicted MCS/rate, and current channel condition. In the implementation, the UE action is used as a scheduling recommendation, while the scheduler module at the gNB produces the final applied decision. If there is no conflict, the applied decision follows the UE recommendation; otherwise, the scheduler modifies the allocation according to the contention-resolution rule. The resulting decision determines the QoS feedback used for DDQN training.

Each UE requests one of {eMBB, URLLC, mMTC} with service-specific throughput, latency, and reliability targets. The target tuple is used both in the reward normalization and in the hard QoS satisfaction metric. A UE is counted as QoS-satisfied only when its throughput, latency, and reliability simultaneously meet the corresponding service targets. The reliability metric is computed as the packet success ratio after BLER- and HARQ-like retransmission modeling within the latency budget.

We adopt the operational time-scale hierarchy defined in Section 3. One RL episode consists of $L = 200$ slots. One FL round is triggered every 5 RL episodes, and VGAE–GMM clustering is refreshed every 10 FL rounds. This schedule reflects the different operating costs of slot-level scheduling, UE-side DDQN training, federated aggregation, and graph-based clustering refresh. We use the same episode, round, exploration, and model-size budgets for all learning-based baselines.

4.2. Performance Metrics and Evaluation Scenarios

We report four primary metrics aligned with the problem objective and the handover/cold-start design goals. First, QoS satisfaction is the fraction of active UEs whose measured throughput, latency, and reliability simultaneously satisfy their service-specific targets. Second, average throughput and latency are computed over the active UE set to evaluate efficiency and delay. Third, reliability is reported separately as the packet success ratio under the BLER and HARQ-like retransmission model, since reliability is an explicit term in the reward and QoS tuple. Fourth, adaptation time measures the elapsed time required for a newly activated or handover UE to recover its service-specific QoS satisfaction after joining or changing cells.

We evaluate TFL-CORAN under a unified scenario that jointly includes cross-cell mobility, new UE activation, and heterogeneous traffic classes. This setting is intended to stress the interaction among dynamic clustering,

Table 4: Simulation and Training Parameters

Category	Setting
Cells / ISD	3 hexagonal cells / 500 m
Initial active UEs	150 (urban), 120 (commercial), 105 (school)
New UE activation	3% of initial UEs per FL round
Handover / reassignment	10% of active UEs per FL round
Population handling	Active UE set grows; no deactivation
Carrier bandwidth	100 MHz at 3.5 GHz
Subbands	$N_f = 6$
Slot / episode	1 ms / 200 slots
FL period	5 RL episodes
Clustering refresh	10 FL rounds
Traffic classes	eMBB, URLLC, mMTC
QoS targets	eMBB: 20 Mbps/15 ms; URLLC: 10 Mbps/5 ms; mMTC: 5 Mbps/10 ms
Reliability model	SINR-to-MCS, BLER, and HARQ-like retransmission
Scheduler rule	Priority/rate-aware contention with predicted MCS
Mobility model	Scenario-dependent bounded random mobility
Urban / commercial / school mobility	High / moderate / low mobility
GCN layers / dimensions	2 / (64, 32)
VGAE latent dimension	32
GMM components	$K = 3$
GMM initialization	k-means first; previous-GMM warm start afterwards
VGAE training data	Historical context-graph snapshots; fixed encoder online
Random seeds	5
Total FL rounds	30

Table 5: Baseline Configurations

Method	**Description**
Heuristic	Priority- and SINR/MCS-based scheduler without learning
DRL	Centralized DDQN policy without FL or clustering
FDRL	FedAvg across UE models without clustering or transfer
CFDRL	Hard-cluster federated DRL with one cluster assignment per UE
TFL–CORAN	Soft-membership aggregation, UE-personalized model synthesis, and transfer initialization

membership-aware aggregation, and transfer initialization. For clustering analysis, we use compactness and separation of the VGAE latent embeddings [30] to support the choice of K and to interpret cluster quality, while end-to-end conclusions are drawn from QoS, throughput, latency, reliability, adaptation time, and overhead results. The computational and communication overhead of UE-side DDQN training and model-update transmission is reported in Section 5.4.

5. Performance Evaluation

This section evaluates TFL–CORAN under the O-RAN simulation setting introduced in Section 4. We focus on end-to-end performance, reliability, empirical convergence behavior, adaptation under mobility and activation events, component analysis, and implementation overhead. All learning-based baselines use the same carrier setting, episode length, optimizer, replay-buffer size, and total training budget. They differ only in how coordination and model aggregation are performed.

5.1. Baselines and Evaluation Protocol

The baselines are Heuristic, DRL, FDRL, and CFDRL. Heuristic uses a non-learning priority/SINR-based scheduler. DRL uses a centralized single DDQN model. FDRL applies FedAvg across UEs without clustering. CFDRL uses

Table 6: Performance Comparison by Cell Type and Traffic Type

Method	Cell type	QoS (%)	Thr (Mbps)	Delay (ms)	Traffic type	QoS (%)	Thr (Mbps)	Delay (ms)
Heuristic	Urban	71.3	18.517	16.501	eMBB	71.0	18.513	16.241
	Commercial	73.1	19.123	14.302	URLLC	73.2	10.108	6.102
	School	75.8	19.209	14.049	mMTC	76.0	5.498	12.003
	Avg	73.4	18.976	15.038	Avg	73.4	11.373	11.448
DRL	Urban	86.5	21.145	12.678	eMBB	86.2	20.504	14.507
	Commercial	86.2	22.143	11.356	URLLC	**87.5**	10.204	5.001
	School	86.1	22.896	11.036	mMTC	85.2	5.801	10.205
	Avg	86.3	22.728	11.426	Avg	86.3	12.836	9.904
FDRL	Urban	82.5	20.089	13.567	eMBB	83.2	19.003	15.503
	Commercial	82.6	21.231	12.995	URLLC	82.2	10.001	5.504
	School	83.1	21.612	12.823	mMTC	82.8	5.603	10.801
	Avg	82.7	21.480	13.244	Avg	82.7	11.536	10.603
CFDRL	Urban	86.9	21.467	13.004	eMBB	87.3	20.801	14.203
	Commercial	86.6	22.597	11.432	URLLC	86.4	10.502	4.902
	School	87.0	22.852	11.197	mMTC	86.8	6.101	9.803
	Avg	86.8	22.305	11.997	Avg	86.8	12.468	9.636
TFL–CORAN	Urban	**87.7**	**23.001**	**10.301**	eMBB	**88.5**	**23.001**	**11.002**
	Commercial	**87.5**	**23.009**	**10.247**	URLLC	87.0	**11.003**	**4.804**
	School	**88.0**	**23.426**	**10.163**	mMTC	**87.5**	**6.203**	**9.502**
	Avg	**87.7**	**23.290**	**10.237**	Avg	**87.7**	**13.402**	**8.436**

Thr = average throughput; Delay = average end-to-end latency. Bold indicates best per row.

hard clustering with per-cluster aggregation. TFL–CORAN differs by combining VGAE-based latent representation, GMM soft clustering, membership-weighted aggregation, and transfer initialization. The RIC is the only aggregation entity; gNBs relay UE updates and enforce feasible scheduling decisions.

5.2. End-to-End Performance and Reliability

Table 6 summarizes the performance by cell type and traffic type. Across all settings, TFL–CORAN achieves the best average QoS satisfaction while also improving throughput and reducing delay. The gains are consistent across Urban, Commercial, and School cells. Although the absolute QoS margin over CFDRL is comparable across cells, the Urban result is particularly important because it corresponds to the most challenging regime with stronger interference and more frequent handovers. Even in this setting, TFL–CORAN reaches 87.7% QoS satisfaction with 23.001 Mbps throughput and 10.301 ms delay.

At the traffic level, the improvement is most visible for eMBB, where QoS satisfaction increases from 87.3% to 88.5% and throughput increases from 20.801 Mbps to 23.001 Mbps compared with CFDRL. For URLLC, TFL–CORAN achieves the lowest delay at 4.804 ms while also slightly improving throughput. For mMTC, the gains persist despite dense low-rate access, indicating robustness to bursty device-level traffic. These results suggest that soft clustering and personalized model synthesis improve resource-control quality under heterogeneous UE conditions.

Compared with CFDRL, the main advantage of TFL–CORAN is that it does not rely on hard cluster assignments. In CFDRL, UEs near cluster boundaries can experience abrupt model switching when their contexts change. TFL–CORAN instead synthesizes UE-personalized models as soft mixtures of cluster models, which reduces negative transfer during mobility and is especially beneficial for throughput-oriented eMBB traffic and for UEs in rapidly changing Urban conditions.

Since reliability is part of the QoS tuple and is also included in the reward formulation, it is evaluated explicitly rather than being left implicit inside the QoS satisfaction metric. Table 7 reports the reliability performance for each traffic type. TFL–CORAN achieves the highest reliability in all three traffic classes, indicating that its improvements

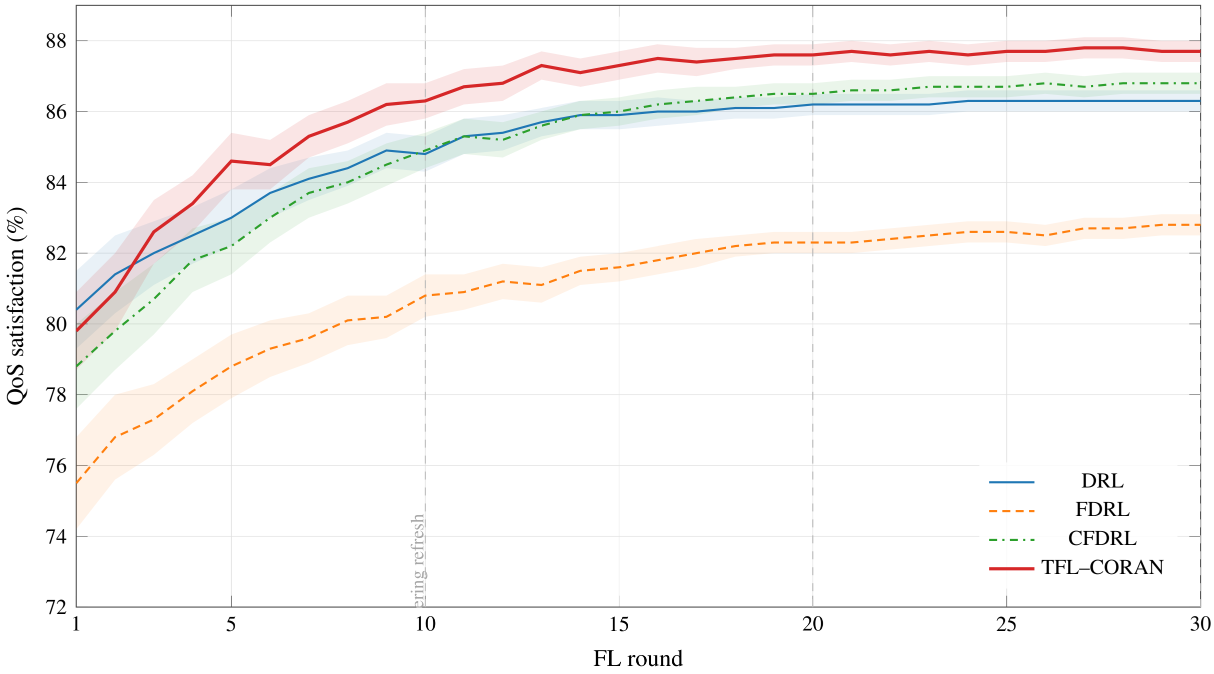

Figure 3: QoS satisfaction over FL rounds. Curves are averaged over five random seeds, and shaded regions indicate standard deviation. Vertical dashed lines denote VGAE–GMM clustering refresh intervals. TFL–CORAN converges faster and exhibits smaller fluctuations than FDRL and CFDRL under the evaluated mobility setting.

Table 7: Reliability Performance

Method	eMBB(%)	URLLC(%)	mMTC(%)
Heuristic	91.8	94.1	90.7
DRL	96.2	98.3	95.1
FDRL	95.4	97.1	94.0
CFDRL	96.8	98.5	96.0
TFL–CORAN	**97.6**	**99.1**	**96.8**

Table 8: Adaptation Time

Method	New UE(s)	Handover UE(s)
Heuristic	–	–
DRL	4.91	4.38
FDRL	5.43	4.87
CFDRL	4.07	3.54
TFL–CORAN	**3.20**	**2.81**

Table 9: Ablation by Component Toggle (O = enabled, X = disabled)

Scheme	TL	VGAE	GMM	QoS (%)	Thr (Mbps)	Adapt. (s)
TFL–CORAN	**O**	**O**	**O**	**87.4**	**23.290**	**3.201**
Variant A	X	O	O	83.0	22.124	5.672
Variant B	X	X	O	82.0	21.932	6.200
Variant C	X	O	X	81.5	21.641	6.012
Variant D	X	X	X	78.0	21.312	5.836

in QoS satisfaction are not obtained by trading away service reliability. This result supports the interpretation that the proposed framework improves not only throughput and latency, but also overall service consistency.

5.3. Convergence, Adaptation, and Component Analysis

Figure 3 shows the empirical learning behavior of the learning-based methods. Rather than claiming formal stability from time-scale separation, we interpret these results as empirical evidence that the proposed time-scale design yields faster convergence and smaller fluctuations than FDRL and CFDRL in the evaluated setting. This

Table 10: Sensitivity to Transfer Fusion Coefficient

δ	QoS(%)	Handover Adapt.(s)
0.00	87.2	3.08
0.25	87.3	2.91
0.50	**87.4**	**2.81**
0.75	87.1	2.95
1.00	86.8	3.32

Table 11: Effect of the Number of GMM Components

K	Compact. ↓	Sep. ↑	QoS(%)
2	0.311	1.418	86.9
3	0.245	1.732	**87.4**
4	0.231	1.754	87.3
5	0.225	1.761	87.1

behavior is consistent with the role of slower clustering refresh and membership-weighted aggregation, which reduce abrupt policy changes while still allowing fast local adaptation at the UE side.

Table 8 reports adaptation times for newly activated and handover UEs. TFL–CORAN reaches target QoS in 3.20 s for new UEs and 2.81 s for handover UEs, outperforming all baselines. This improvement comes from transfer initialization and subsequent local refinement. New UEs receive an informed starting point instead of a generic initialization, while handover UEs benefit from transferring prior knowledge together with target-context alignment.

Table 9 isolates the effects of transfer learning, VGAE embeddings, and GMM clustering. Removing transfer learning primarily increases adaptation time, confirming its importance for cold-start mitigation. Removing VGAE or GMM reduces QoS and throughput while also slowing adaptation, because the RIC loses either graph-informed representation or soft context membership. The full configuration provides the best overall trade-off, showing that the three components play complementary roles rather than redundant ones.

We also examine two parameters related to adaptation and clustering: the transfer fusion coefficient δ and the number of GMM components K. For handover UEs, δ balances the previous-cell model and the target-context model. Table 10 shows that an intermediate value provides the best adaptation trade-off; therefore, we use $\delta = 0.5$ in the main evaluation. For clustering, Table 11 shows that $K = 3$ provides a balanced trade-off between compactness, separation, and QoS performance. Increasing K beyond 3 marginally improves compactness but does not improve end-to-end QoS, likely because excessive fragmentation reduces the amount of useful update sharing within each cluster.

5.4. Computational and Communication Complexity

We compare the computational and communication overhead of TFL-CORAN with the baselines using hardware-independent complexity terms. This is more appropriate than reporting only measured execution time, since actual latency depends on UE chipset, RIC server resources, software implementation, and parallelization. Let N denote the number of active UEs, N_g the number of UEs served by a gNB, N_f the number of subbands, P_Q the number of DDQN parameters, G_{UE} the number of local gradient steps per FL round, B the mini-batch size, d_x the dimension of the context report, d_z the VGAE latent dimension, $\bar{d}$ the average graph degree, K the number of clusters, I_{EM} the number of GMM EM iterations, and τ_C the clustering refresh period in FL rounds.

Table 12 compares the main complexity terms. The Heuristic baseline has the lowest computational cost because it does not train learning models, but it also lacks adaptation from experience. DRL avoids FL communication but requires centralized state collection and centralized training. FDRL adds model-update communication and RIC-side FedAvg aggregation. CFDRL further adds hard clustering overhead. TFL-CORAN has additional VGAE–GMM inference and membership-aware aggregation costs; however, these structural operations are performed only every τ_C FL rounds and are therefore amortized over multiple local training rounds.

Here, d_s is the dimension of the centralized state report, G_c is the number of centralized DDQN gradient steps, I_K is the number of hard-clustering iterations in CFDRL, and C_{GCN} denotes the fixed per-edge cost of the GCN encoder. For sparse graph construction, TFL-CORAN uses approximate neighbor search, yielding $O(N \log N + N\bar{d}d_x)$ graph-construction cost; full pairwise construction would instead require $O(N^2 d_x)$. The table shows that TFL-CORAN is more computationally involved than FDRL and CFDRL because it adds VGAE–GMM-based soft context modeling. However, this overhead is incurred at the slower clustering-refresh scale and is amortized by τ_C. Moreover, the UE-side training order is the same as FDRL and CFDRL, and the per-UE model-update payload remains $O(P_Q)$ per FL round. Thus, the additional cost of TFL-CORAN is concentrated at the RIC-side coordinator, where it is traded for improved personalization, faster adaptation, and reduced negative transfer under mobility-driven heterogeneity.

Table 12: Complexity Comparison of Baselines and TFL-CORAN

Method	Local compute	Coordinator compute	Communication
Heuristic	gNB scheduling: $O(N_g N_f)$ per slot	None	None
DRL	Centralized DDQN training: $O(G_c B P_Q)$	Central controller	State collection: $O(N d_s)$
FDRL	UE local DDQN: $O(N G_{\text{UE}} B P_Q)$	FedAvg: $O(N P_Q)$	Model exchange: $O(N P_Q)$
CFDRL	UE local DDQN: $O(N G_{\text{UE}} B P_Q)$	Hard clustering: $O(I_K N K d_x)$; aggregation: $O(N P_Q)$	Model exchange: $O(N P_Q)$; context reports: $O(N d_x)$
TFL–CORAN	UE local DDQN: $O(N G_{\text{UE}} B P_Q)$	Aggregation/personalization: $O(N K P_Q)$; amortized VGAE–GMM: $\frac{1}{\tau_C} O(N \log N + N \bar{d} C_{\text{GCN}} + I_{\text{EM}} K N d_z^2)$	Model exchange: $O(N P_Q)$; amortized context reports: $\frac{1}{\tau_C} O(N d_x)$

Note: N is the number of active UEs, N_g is the number of UEs served by a gNB, N_f is the number of subbands, P_Q is the number of DDQN parameters, G_{UE} is the number of local gradient steps, B is the mini-batch size, d_x is the context dimension, d_z is the latent dimension, K is the number of clusters, and τ_C is the clustering refresh period.

6. Conclusion

We presented TFL-CORAN, a UE-centric federated reinforcement learning framework for traffic management in 5G O-RAN. The proposed framework explicitly separates the roles of UEs, gNBs, and the RIC: UEs perform local DDQN-based learning, gNBs enforce feasible slot-level scheduling and relay model/context information, and the RIC coordinates representation learning, soft clustering, federated aggregation, and personalized model redistribution. To address mobility-driven non-IID behavior and cold starts, TFL-CORAN combines offline VGAE-based context embedding, online GMM soft membership inference, membership-weighted federated aggregation, and transfer-based initialization for newly activated and handover UEs. This design enables UE-level personalization without centralized raw data collection and avoids expensive online graph retraining.

Simulation results show that TFL-CORAN improves QoS satisfaction, throughput, latency, reliability, and adaptation time compared with centralized DRL, conventional federated DRL, and hard-cluster federated DRL baselines. In the evaluated O-RAN scenarios, TFL-CORAN achieves an average QoS satisfaction of approximately 87.7%, average throughput of 23.29 Mbps, and average delay of 10.24 ms across cell types. It also shortens adaptation time to 3.20 s for newly activated UEs and 2.81 s for handover UEs, demonstrating the benefit of transfer-based warm starting. The ablation and sensitivity results further confirm that VGAE representation learning, GMM soft clustering, and transfer initialization provide complementary benefits rather than redundant improvements.

While the current evaluation is simulation-based and considers a controlled set of O-RAN scenarios, the results provide empirical evidence that time-scale-separated clustering, membership-aware aggregation, and transfer initialization can improve UE-level adaptation under mobility-driven heterogeneity. Future extensions include larger-scale multi-vendor testbed validation, richer cross-cell coordination among RIC functions, and further reduction of UE-side computation and communication overhead through lightweight inference, update compression, and efficiency-aware scheduling. Overall, TFL-CORAN provides a coherent learning-and-control pipeline for scalable, mobility-aware, and personalized traffic management in 5G O-RAN.

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